DATABASE RELATIONSHIP ANALYSIS USING JOINS

PURPLLE ONLINE BEAUTY SHOPPING

Introduction

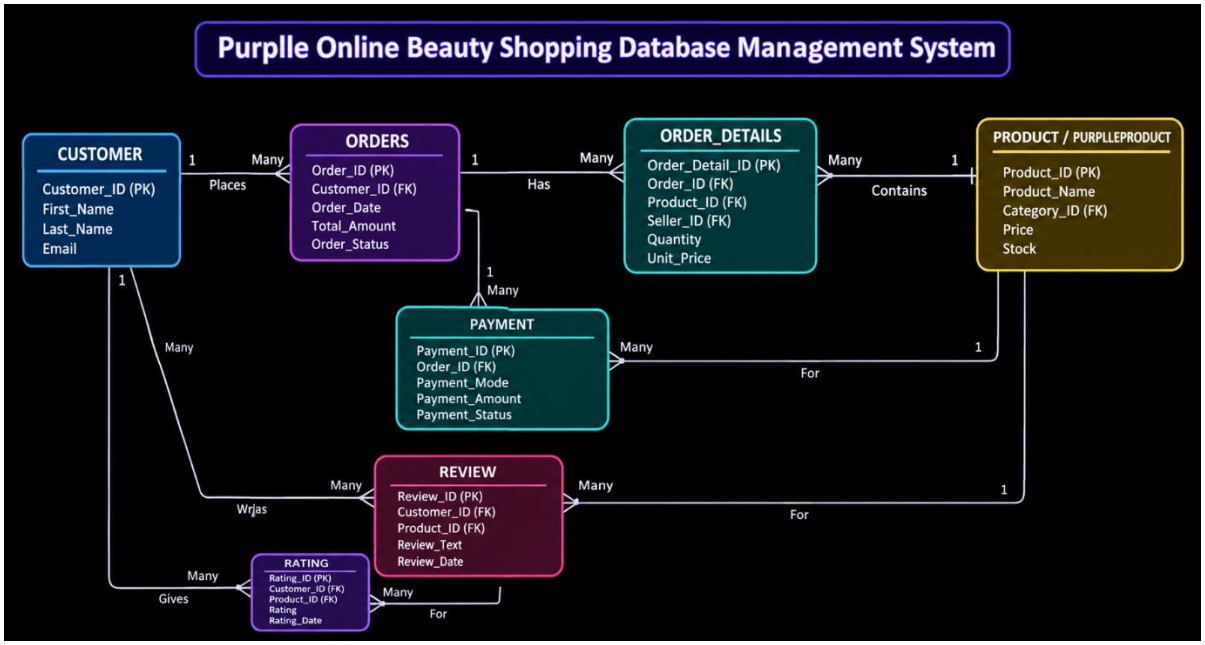
The Purplle Online Beauty Shopping Database Management System uses SQL JOIN operations to combine related information from multiple tables. This week focuses on INNER JOIN, LEFT JOIN, and RIGHT JOIN to retrieve customer, product, order, and payment details. SQL queries are used to display customer purchase histories and order information. Aggregate functions such as COUNT() and SUM() help generate business reports. These operations improve data retrieval and support efficient database management.

Objectives

1. Combine Customer, Product, Order, and Payment tables.
2. Implement INNER JOIN, LEFT JOIN, and RIGHT JOIN.
3. Retrieve complete order details.
4. Display customer purchase history.
5. Generate multi-table reports.

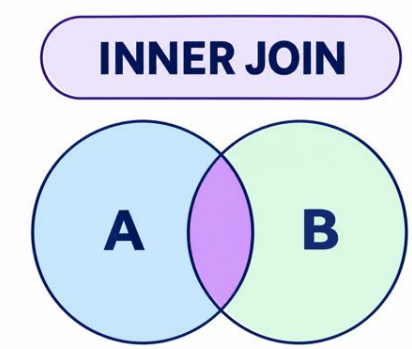
1.Combining Multiple Tables

The EcoCart database contains different tables that store customer, product, order, and payment information. The tables can be combined using JOIN operations to retrieve related information.



2. INNER JOIN

INNER JOIN returns only the records that have matching values in both tables.



2.1 Customer and Orders

This displays customers who have placed orders.

```
SQL> SELECT
  2       C.Customer_ID,
  3       C.First_Name,
  4       C.Last_Name,
  5       O.Order_ID,
  6       O.Order_Date,
  7       O.Total_Amount,
  8       O.Order_Status
  9  FROM Customer5 C
 10  INNER JOIN Orders O
 11       ON C.Customer_ID = O.Customer_ID
 12  ;

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                          ORDER_ID ORDER_DAT
-------------------------------------------------- ---------- ---------
TOTAL_AMOUNT ORDER_STATUS
------------ ------------------------------
         101 Nikitha
R                                                         501 20-SEP-26
         1699 Pending

         102 Priya
Sharma                                                    502 26-SEP-26
          749 Shipped

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                          ORDER_ID ORDER_DAT
-------------------------------------------------- ---------- ---------
TOTAL_AMOUNT ORDER_STATUS
------------ ------------------------------

         103 Arun
Kumar                                                     503 26-SEP-26
         1299 Delivered

         104 Rahul
Verma                                                     504 26-SEP-26

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                          ORDER_ID ORDER_DAT
-------------------------------------------------- ---------- ---------
TOTAL_AMOUNT ORDER_STATUS
------------ ------------------------------
          799 Pending

         105 Sneha
Patel                                                     505 26-SEP-26
         1599 Delivered
```

2.2 Orders and Payment

This displays order and corresponding payment details.

```
SQL> SELECT
  2     O.Order_ID,
  3     O.Order_Date,
  4     O.Total_Amount,
  5    P.Payment_Mode,
  6     P.Payment_Amount,
  7     P.Payment_Status
  8  FROM Orders O INNER JOIN Payment P ON O.Order_ID = P.Order_ID;

  ORDER_ID ORDER_DAT TOTAL_AMOUNT PAYMENT_MODE         PAYMENT_AMOUNT
---------- --------- ------------ -------------------- --------------
PAYMENT_STATUS
--------------------
       501 20-SEP-26         1699 UPI                            1699
Successful

       502 26-SEP-26          749 CARD                            749
Successful

       503 26-SEP-26         1299 COD                            1299
Successful


  ORDER_ID ORDER_DAT TOTAL_AMOUNT PAYMENT_MODE         PAYMENT_AMOUNT
---------- --------- ------------ -------------------- --------------
PAYMENT_STATUS
--------------------
       504 26-SEP-26          799 UPI                             799
Successful

       505 26-SEP-26         1599 CARD                           1599
Successful
```

2.3 Product and Order_Details

This displays products that have been included in orders.

```
SQL> SELECT
  2      P.Product_ID,
  3      P.Product_Name,
  4      OD.Order_ID,
  5      OD.Quantity,
  6      OD.Unit_Price,
  7      (OD.Quantity * OD.Unit_Price) AS Subtotal
  8  FROM PurppleProduct P
  9  INNER JOIN Order_Details OD
 10      ON P.Product_ID = OD.Product_ID
 11  ;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
  ORDER_ID   QUANTITY UNIT_PRICE   SUBTOTAL
---------- ---------- ---------- ----------
       101
Lakme Foundation
       501          3        799       2397

       102
Maybelline Lipstick
       502          1        749        749

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
  ORDER_ID   QUANTITY UNIT_PRICE   SUBTOTAL
---------- ---------- ---------- ----------

       103
Mamaearth Face Wash
       503          2      649.5       1299

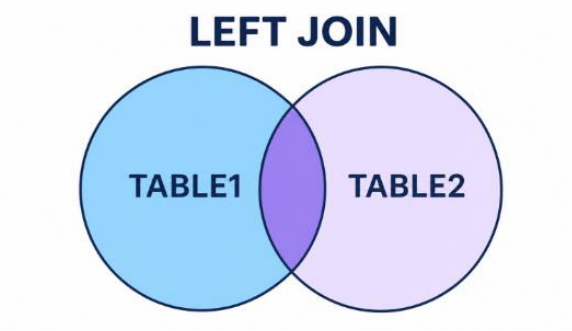
       104
Minimalist Serum

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
  ORDER_ID   QUANTITY UNIT_PRICE   SUBTOTAL
---------- ---------- ---------- ----------
       504          1        799        799

       105
Loreal Shampoo
       505          2      799.5       1599
```

3. LEFT JOIN

LEFT JOIN returns all records from the left table and matching records from the right table.



3.1 Display All Customers and Their Orders

This displays all customers, including customers who have not placed an order.

```
SQL> SELECT
  2      C.Customer_ID,
  3      C.First_Name,
  4      C.Last_Name,
  5      O.Order_ID,
  6      O.Order_Date,
  7      O.Total_Amount
  8  FROM Customer5 C
  9  LEFT JOIN Orders O
 10      ON C.Customer_ID = O.Customer_ID;

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                          ORDER_ID ORDER_DAT
-------------------------------------------------- ---------- ---------
TOTAL_AMOUNT
------------
        101 Nikitha
R                                                         501 20-SEP-26
        1699

        102 Priya
Sharma                                                    502 26-SEP-26
         749

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                          ORDER_ID ORDER_DAT
-------------------------------------------------- ---------- ---------
TOTAL_AMOUNT
------------

        103 Arun
Kumar                                                     503 26-SEP-26
        1299

        104 Rahul
Verma                                                     504 26-SEP-26

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                          ORDER_ID ORDER_DAT
-------------------------------------------------- ---------- ---------
TOTAL_AMOUNT
------------
         799

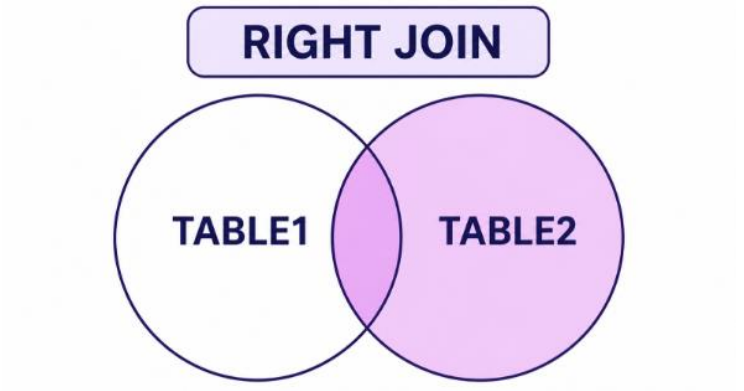
        105 Sneha
Patel                                                     505 26-SEP-26
        1599

        107 Karthik

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                          ORDER_ID ORDER_DAT
-------------------------------------------------- ---------- ---------
TOTAL_AMOUNT
------------
```

4. RIGHT JOIN

RIGHT JOIN returns all records from the right table and matching records from the left table.



4.1 Display All Orders and Customer Details

This displays all orders along with the matching customer information.

```
SQL> SELECT
  2  C.Customer_ID,
  3  C.First_Name,
  4  O.Order_ID,
  5  O.Order_Date,
  6  O.Total_Amount
  7  FROM Customer5 C
  8  RIGHT JOIN Orders O
  9  ON C.Customer_ID = O.Customer_ID;

CUSTOMER_ID FIRST_NAME                                             ORDER_ID
----------- -------------------------------------------------- ----------
ORDER_DAT TOTAL_AMOUNT
--------- ------------
        101 Nikitha                                                   501
20-SEP-26         1699

        102 Priya                                                     502
26-SEP-26          749

        103 Arun                                                      503
26-SEP-26         1299


CUSTOMER_ID FIRST_NAME                                             ORDER_ID
----------- -------------------------------------------------- ----------
ORDER_DAT TOTAL_AMOUNT
--------- ------------
        104 Rahul                                                     504
26-SEP-26          799

        105 Sneha                                                     505
26-SEP-26         1599
```

5. Complete Order Details

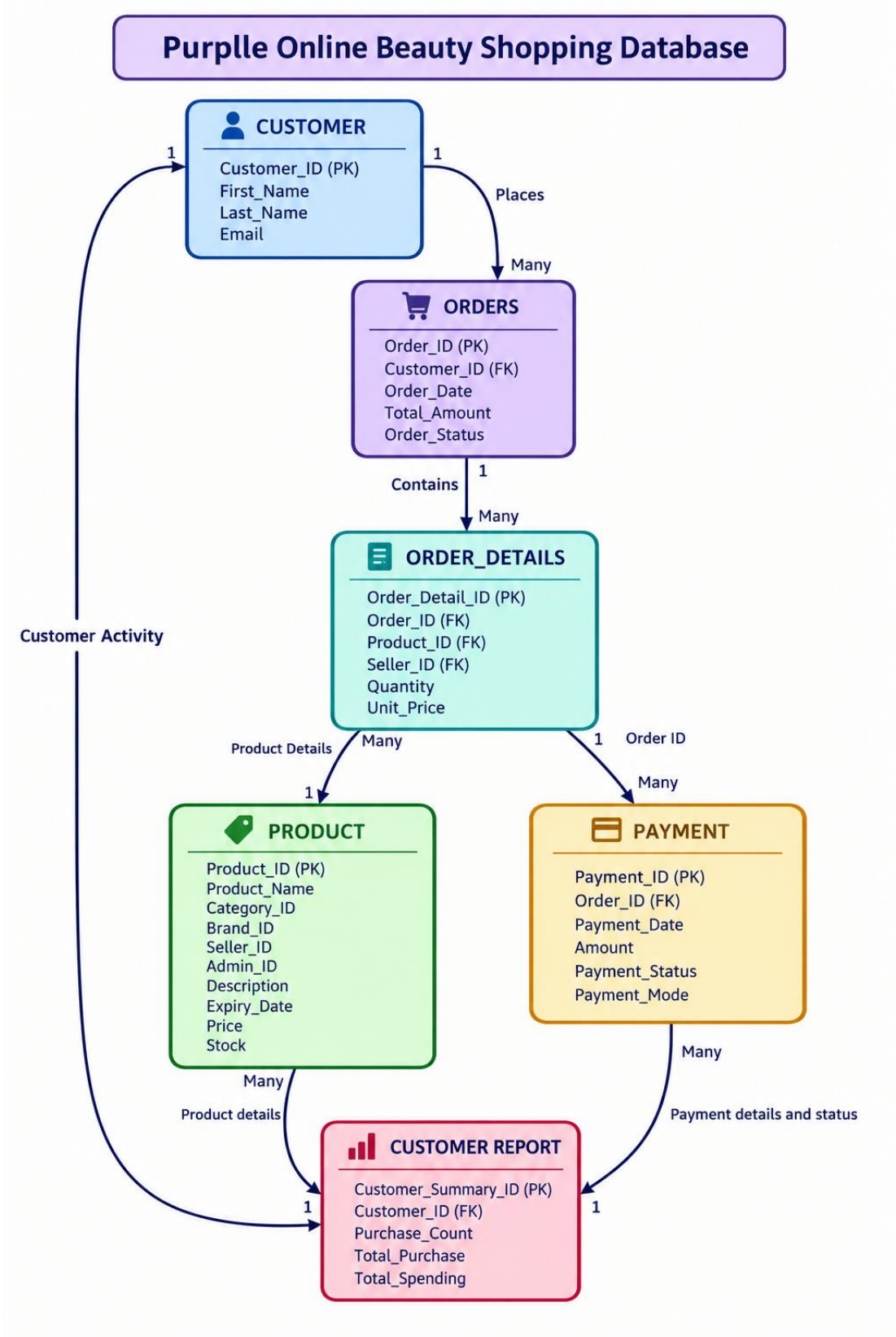
This query combines Customer, Orders, Order_Details, Product, and Payment tables.

```
QL> SELECT
 2      C.Customer_ID,
 3      C.First_Name,
 4      C.Last_Name,
 5      O.Order_ID,
 6      O.Order_Date,
 7      P.Product_ID,
 8      P.Product_Name,
 9      OD.Quantity,
10      OD.Unit_Price,
11      (OD.Quantity * OD.Unit_Price) AS Subtotal,
12      PY.Payment_Mode,
13      PY.Payment_Amount,
14      PY.Payment_Status
15  FROM Customer5 C
16  INNER JOIN Orders O
17      ON C.Customer_ID = O.Customer_ID
18  INNER JOIN Order_Details OD
19      ON O.Order_ID = OD.Order_ID
20  INNER JOIN PurplleProduct P
21      ON OD.Product_ID = P.Product_ID
22  INNER JOIN Payment PY
23      ON O.Order_ID = PY.Order_ID
24  ORDER BY O.Order_Date DESC;

USTOMER_ID FIRST_NAME
---------- --------------------------------------------------
AST_NAME                                           ORDER_ID ORDER_DAT
-------------------------------------------------- ---------- ---------
RODUCT_ID
---------
RODUCT_NAME
--------------------------------------------------------------------------------
 QUANTITY UNIT_PRICE   SUBTOTAL PAYMENT_MODE         PAYMENT_AMOUNT
--------- ---------- ---------- -------------------- --------------
AYMENT_STATUS
--------------------
       105 Sneha

USTOMER_ID FIRST_NAME
---------- --------------------------------------------------
AST_NAME                                           ORDER_ID ORDER_DAT
-------------------------------------------------- ---------- ---------
RODUCT_ID
---------
RODUCT_NAME
--------------------------------------------------------------------------------
 QUANTITY UNIT_PRICE   SUBTOTAL PAYMENT_MODE         PAYMENT_AMOUNT
--------- ---------- ---------- -------------------- --------------
AYMENT_STATUS
--------------------
atel                                                    505 26-SEP-26

USTOMER_ID FIRST_NAME
---------- --------------------------------------------------
AST_NAME                                           ORDER_ID ORDER_DAT
-------------------------------------------------- ---------- ---------
RODUCT_ID
```


Purplle Online Beauty Shopping Database
CUSTOMER
Customer_ID (PK)
First_Name
Last_Name
Email
1
1
Places
Many
ORDERS
Order_ID (PK)
Customer_ID (FK)
Order_Date
Total_Amount
Order_Status
Contains
1
Many
ORDER_DETAILS
Order_Detail_ID (PK)
Order_ID (FK)
Product_ID (FK)
Seller_ID (FK)
Quantity
Unit_Price
Customer Activity
Product Details
Many
1
1
Order ID
Many
PRODUCT
Product_ID (PK)
Product_Name
Category_ID
Brand_ID
Seller_ID
Admin_ID
Description
Expiry_Date
Price
Stock
PAYMENT
Payment_ID (PK)
Order_ID (FK)
Payment_Date
Amount
Payment_Status
Payment_Mode
Many
Product details
Payment details and status
CUSTOMER REPORT
Customer_Summary_ID (PK)
Customer_ID (FK)
Purchase_Count
Total_Purchase
Total_Spending
1
1

6. Customer Purchase History

This query displays the products purchased by each customer.

```
SQL> SELECT
  2       C.Customer_ID,
  3       C.First_Name,
  4       C.Last_Name,
  5       O.Order_ID,
  6       P.Product_Name,
  7       OD.Quantity,
  8       (OD.Quantity * OD.Unit_Price) AS Subtotal
  9  FROM Customer5 C
 10  INNER JOIN Orders O
 11       ON C.Customer_ID = O.Customer_ID
 12  INNER JOIN Order_Details OD
 13       ON O.Order_ID = OD.Order_ID
 14  INNER JOIN PurplleProduct P
 15       ON OD.Product_ID = P.Product_ID
 16  ORDER BY C.Customer_ID, O.Order_Date;

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                           ORDER_ID
-------------------------------------------------- ----------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY   SUBTOTAL
---------- ----------
         101 Nikitha
R                                                         501
Lakme Foundation
         3       2397


CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                           ORDER_ID
-------------------------------------------------- ----------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY   SUBTOTAL
---------- ----------
         102 Priya
Sharma                                                    502
Maybelline Lipstick
         1        749


CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                           ORDER_ID
-------------------------------------------------- ----------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY   SUBTOTAL
---------- ----------
         103 Arun
Kumar                                                     503
Mamaearth Face Wash
         2       1299


CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                           ORDER_ID
-------------------------------------------------- ----------
```

7. Customer Purchase History with Payment

```
SQL> SELECT
  2      C.Customer_ID,
  3      C.First_Name,
  4      C.Last_Name,
  5      O.Order_ID,
  6      O.Order_Date,
  7      P.Product_Name,
  8      OD.Quantity,
  9      (OD.Quantity * OD.Unit_Price) AS Subtotal,
 10      PY.Payment_Mode,
 11      PY.Payment_Status
 12  FROM Customer5 C
 13  INNER JOIN Orders O
 14      ON C.Customer_ID = O.Customer_ID
 15  INNER JOIN Order_Details OD
 16      ON O.Order_ID = OD.Order_ID
 17  INNER JOIN PurplleProduct P
 18      ON OD.Product_ID = P.Product_ID
 19  INNER JOIN Payment PY
 20      ON O.Order_ID = PY.Order_ID
 21  ORDER BY C.Customer_ID;

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                          ORDER_ID ORDER_DAT
-------------------------------------------------- ---------- ---------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY   SUBTOTAL PAYMENT_MODE         PAYMENT_STATUS
---------- ---------- -------------------- --------------------
         101 Nikitha
R                                                     501 20-SEP-26
Lakme Foundation
         3       2397 UPI                  Successful


CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                          ORDER_ID ORDER_DAT
-------------------------------------------------- ---------- ---------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY   SUBTOTAL PAYMENT_MODE         PAYMENT_STATUS
---------- ---------- -------------------- --------------------
         102 Priya
Sharma                                                502 26-SEP-26
Maybelline Lipstick
         1        749 CARD                 Successful


CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                          ORDER_ID ORDER_DAT
-------------------------------------------------- ---------- ---------
```

```
---------- ---------- -------------------- --------------------
        102 Priya
Sharma                                                 502 26-SEP-26
Maybelline Lipstick
         1        749 CARD                 Successful


CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                          ORDER_ID ORDER_DAT
-------------------------------------------------- ---------- ---------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY   SUBTOTAL PAYMENT_MODE         PAYMENT_STATUS
---------- ---------- -------------------- --------------------
        103 Arun
Kumar                                                  503 26-SEP-26
Mamaearth Face Wash
         2       1299 COD                  Successful


CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                          ORDER_ID ORDER_DAT
-------------------------------------------------- ---------- ---------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY   SUBTOTAL PAYMENT_MODE         PAYMENT_STATUS
---------- ---------- -------------------- --------------------
        104 Rahul
Verma                                                  504 26-SEP-26
Minimalist Serum
         1        799 UPI                  Successful


CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                          ORDER_ID ORDER_DAT
-------------------------------------------------- ---------- ---------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY   SUBTOTAL PAYMENT_MODE         PAYMENT_STATUS
---------- ---------- -------------------- --------------------
        105 Sneha
Patel                                                  505 26-SEP-26
Loreal Shampoo
         2       1599 CARD                 Successful
```

8. Multi-Table Business Report

The following query generates a complete customer purchase report.

```
SQL> SELECT
  2      C.Customer_ID,
  3      C.First_Name || ' ' || C.Last_Name AS Customer_Name,
  4      COUNT(DISTINCT O.Order_ID) AS Total_Orders,
  5      SUM(OD.Quantity * OD.Unit_Price) AS Total_Purchase,
  6      SUM(OD.Quantity) AS Total_Products
  7  FROM Customer5 C
  8  INNER JOIN Orders O
  9      ON C.Customer_ID = O.Customer_ID
 10  INNER JOIN Order_Details OD
 11      ON O.Order_ID = OD.Order_ID
 12  GROUP BY
 13      C.Customer_ID,
 14      C.First_Name,
 15      C.Last_Name
 16  ORDER BY Total_Purchase DESC;

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
TOTAL_ORDERS TOTAL_PURCHASE TOTAL_PRODUCTS
------------ -------------- --------------
        101
Nikitha R
           1           2397              3

        105
Sneha Patel
           1           1599              2

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
TOTAL_ORDERS TOTAL_PURCHASE TOTAL_PRODUCTS
------------ -------------- --------------

        103
Arun Kumar
           1           1299              2

        104
Rahul Verma

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
TOTAL_ORDERS TOTAL_PURCHASE TOTAL_PRODUCTS
------------ -------------- --------------
           1            799              1

        102
Priya Sharma
           1            749              1
```

9. Payment Transaction Report

```
SQL> SELECT
  2  C.First_nAME,
  3  O.Order_ID,
  4  O.Order_Date,
  5  O.Total_Amount,
  6  PY.Payment_Mode,
  7  PY.Payment_Amount,
  8  PY.Payment_Status
  9  FROM Customer5 C
 10  INNER JOIN Orders O
 11      ON C.Customer_ID = O.Customer_ID
 12  INNER JOIN PAYMENT PY
 13      ON O.Order_ID = PY.Order_ID
 14  ORDER BY O.Order_Date DESC;

FIRST_NAME                                         ORDER_ID ORDER_DAT
-------------------------------------------------- ---------- ---------
TOTAL_AMOUNT PAYMENT_MODE         PAYMENT_AMOUNT PAYMENT_STATUS
------------ -------------------- -------------- --------------------
Sneha                                                   505 26-SEP-26
        1599 CARD                           1599 Successful

Rahul                                                   504 26-SEP-26
         799 UPI                             799 Successful

Arun                                                    503 26-SEP-26
        1299 COD                            1299 Successful


FIRST_NAME                                         ORDER_ID ORDER_DAT
-------------------------------------------------- ---------- ---------
TOTAL_AMOUNT PAYMENT_MODE         PAYMENT_AMOUNT PAYMENT_STATUS
------------ -------------------- -------------- --------------------
Priya                                                   502 26-SEP-26
         749 CARD                            749 Successful

Nikitha                                                 501 20-SEP-26
        1699 UPI                            1699 Successful
```

Conclusion

The database relationship analysis was completed using SQL JOIN operations. Customer, product, order, and payment details were retrieved by combining multiple tables. Different JOIN types helped display matching records and related information. Aggregate functions were used to calculate total orders, purchase amounts, and product quantities. This task improved understanding of table relationships and multi-table queries in SQL.